

ZHIHENG LYU

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PROFILE

Incoming CS PhD at UIUC (advisor: **Lingming Zhang**), specializing in LLM post-training for code and SWE agents. Production experience shipping MiniMax M1–M2.5. Core developer of VerlTool. ICPC World Finalist.

EDUCATION

University of Illinois Urbana-Champaign Ph.D. in Computer Science Supervised by Prof. Lingming Zhang	2026 Fall (Incoming) <i>Urbana-Champaign, IL, USA</i>
University of Waterloo M.Math. in Computer Science Supervised by Prof. Wenhui Chen	Sept. 2024 – June 2026 (Expected) <i>Waterloo, ON, Canada</i>
The University of Hong Kong B.Eng. in Computer Science, First Class Honours degree Entered 45th ICPC World Final; Got Two Gold Medals in ICPC Regional (Nanjing & Shanghai);	2020 – 2024 <i>Hong Kong SAR, China</i>
University of California, Berkeley Visiting Student for EECS Department	2022 – 2023 <i>Berkeley, CA, USA</i>

WORK EXPERIENCE

MiniMax Research Scientist Intern at Base Model Team - Contributed to MiniMax M1, M2, M2.1, and M2.5 code-agent post-training under the supervision of Qidi Wu and Pengyu Zhao. Designed an agentic RL framework based on VerlTool for SWE-Bench, outperforming SkyRL on Qwen3-8B. M2 achieved 69% Pass@1 on SWE-Verified, ranking #2 on MultiSWE and TerminalBench. For M2.5, led CodeMirror ToolScaling: clustered tool-use patterns and expanded by category, enabling the model to generalize across diverse scaffolding frameworks. Led large-scale SWE data synthesis: generated 36K verifiable tasks from 5K+ sandbox environments, rewrote 7K+ non-verifiable tasks from GitHub Issues and Cursor queries, and applied rule-based cleaning and generative reward modeling (GRM) for quality filtering. Collaborated with infra to resolve sandbox bottlenecks, improving throughput by 80%. - Built an internal rubric-based evaluation benchmark from live user feedback and return traces (~100B tokens/day) for M2.1, becoming the core metric for diagnosing Code-Agent user experience. Enhanced performance through rubric filtering, synthesis, and rewrite.	May 2025 – Feb. 2026 <i>Shanghai, China</i>
Tencent Intern (Qingyun Program), Code/WorkBuddy Agent Post-Training - Worked closely with Ke Li and Chao Peng on WorkBuddy. Owned the team's evaluation platform and infrastructure: automated GPU resource management and resolved sandbox reliability issues, leading to a 3× increase in evaluation volume across multiple agentic benchmarks. - Explored task synthesis from production user data for post-training, including sandbox synthesis from return traces and trajectory rewriting from user interaction logs.	Feb. 2026 – July 2026 <i>Shenzhen, China</i>
Capital Today Group Private Equity Research Intern Conduct private equity investment research for LLMs, Embodied AI, and AI Applications.	May 2024 – Aug. 2024 <i>Shanghai, China</i>

RESEARCH EXPERIENCE

M.Math.(Fully Funded), UWaterloo TIGER Lab <i>Advisor: Prof. Wenhui Chen</i> Thesis: Post-Training LLMs as Software Engineering Agents. Core developer of VerlTool [5] (stateful environment protocol & SWE agent pipeline); co-first authored SWE-Next [2] (2,308 verifiable tasks from 311 repos), SWE-QA-Pro [3] (ACL 2026 Findings; 8B surpasses GPT-4o), and BrowserAgent [4] (TMLR; +17% via SFT→RFT). RLVR on SWE-bench Verified: Qwen3-8B 10.4% → 19.5% (+87.5% relative).	Sept. 2024 – June 2026 <i>Waterloo, Canada</i>
Research Intern, UC Berkeley BAIR, NLP Group <i>Advisors: Prof. Dan Klein & Dr. Kevin Yang</i> Proposed FactTrack [8], decomposing narratives into atomic facts with dynamic world-state tracking for contradiction detection. (NAACL 2025 Oral)	Apr. 2023 – Aug. 2023 <i>Berkeley, USA</i>
Research Intern, ETH Zürich NLPED & University of Michigan LIT <i>Advisors: Prof. Mrinmaya Sachan, Prof. Rada Mihalcea, & Dr. Zhijiang Jin</i>	Jun 2022 – Oct. 2022 <i>Zürich, Switzerland</i>

- Designed the **Corr2Cause** benchmark for evaluating causal inference in LLMs (ICLR 2024) and investigated causal prompting strategies (EMNLP 2024).

SELECTED PUBLICATIONS

1. Tencent WorkBuddy Bench Team*, S. Cai*, S. Chen*, X. Fei*, Y. Mao*, Z. Xu*, **Zhiheng Lyu***, et al. (2026). (Technical Report) Tencent WorkBuddy Bench: A Multi-Domain Coding-Agent Benchmark with Contamination-Resistant Task Construction.
2. J. Liang*, **Zhiheng Lyu***, S. Cai, Y. Lu, P. Nie, W. Chen (2026). (Under Review) SWE-Next: Scaling SWE Task Synthesis for Agentic LLM Training.
3. S. Cai*, **Zhiheng Lyu***, Y. Ni, X. Chen, B. Zhou, S. Zhu, Y. Lu, H. Wang, C. Ruan, B. Schneider, W. Zhang, X. Li, A. Zheng, Y. Zhang, P. Nie, W. Chen (2026). SWE-QA-Pro: A Representative Benchmark and Scalable Training Recipe for Repository-Level Code Understanding.
4. T. Yu*, Z. Zhang*, **Zhiheng Lyu***, J. Gong, H. Yi, X. Wang, Y. Zhou, J. Yang, P. Nie, Y. Huang, W. Chen (2025). (TMLR) BrowserAgent: Building Web Agents with Human-Inspired Web Browsing Actions.
5. D. Jiang*, Z. Li*, Y. Lu*, **Zhiheng Lyu***, P. Nie, W. Chen (2025). VerlTool: A unified and easy-to-extend tool-agent training framework based on verl. (Open Source Project)
6. MiniMax. (2025). MiniMax-M1: Scaling Test-Time Compute Efficiently with Lightning Attention. (Technical Report)
7. **Zhiheng Lyu**, X. Ma, W. Chen (2025). (TMLR) *PixelWorld: Towards Perceiving Everything as Pixels*.
8. **Zhiheng Lyu**, K. Yang, L. Kong, D. Klein (NAACL 2025 Oral). FACTTRACK: Time-Aware World State Tracking in Story Outlines.
9. Plus 7 additional publications at ICLR, NeurIPS, EMNLP, UAI, etc. Full list at <http://cogito233.github.io>.

AWARDS & HONORS

1. Ranked 50 in **45th International Collegiate Programming Contest (ICPC) World Finals**, Dhaka, Bangladesh, Nov. 2022
2. 9th Place Team in **ICPC World Challenge for Real World Tasks**, Dhaka, Bangladesh, Nov. 2022
3. Gold Medalist and Team Captain in ICPC 2021 Shanghai region – ranked 20/668, Shanghai, China, Dec. 2021
4. Gold Medalist and Team Captain in ICPC 2020 Nanjing region – ranked 7/556, Nanjing, China, Dec. 2020
5. Silver Medalist at National Olympiad in Informatics (NOI) 2018, Changsha, China, Jul. 2018
6. Gold Medalist at The Asia-Pacific Informatics Olympiad (APIO) 2018, Beijing, China, Apr. 2018

PROFESSIONAL SERVICE

Reviewer: ICLR 2026, NeurIPS 2025, ICML 2025, ACL 2024.

Co-organizer: Half-day workshop *Performance Optimization in the LLM World* @ ICPE 2024 & 2025.

LEADERSHIP EXPERIENCE

HKU AI4Good HKU, *Founder*

AStar Team @ HKU, *Team Leader*

Sept 2023 – Jun 2024

July 2021 – Aug 2022

SKILLS

Languages	English, Mandarin, Cantonese, Shanghaiese, Classical Chinese
Programming	Python, C/C++, PyTorch, verl, Docker, Distributed Systems, Algorithms & Data Structures
Research	LLM Post-Training, RL for Agents, Software Engineering, SWE-Bench, NLP